

C.H.U.D.: Catastrophic Harmful Update Detection in Sequential Fine-Tuning

Jorge Manuel Torre
Virginia Tech

Elias Mengistu
Virginia Tech

Alex Levy
Virginia Tech

Jeevan Shahi
Virginia Tech

Abstract

Safety alignment in open-weight language models is not permanent: fine-tuning can erode refusal behavior and enable harmful outputs. We investigate whether catastrophic forgetting can be exploited in a sequential fine-tuning attack (first on benign data, then on harmful data) and whether LoX (Low-Rank Extrapolation), a recent defense against fine-tuning attacks, holds up under this threat model. Using our evaluation pipeline C.H.U.D. (Catastrophic Harmful Update Detection), we fine-tune Llama-2-7B-Chat on GSM8K math data followed by BeaverTails harmful samples, then measure Attack Success Rate (ASR) on AdvBench prompts using Llama-Guard-3-8B as a judge. Under direct harmful fine-tuning, LoX reduces ASR from 91% to 37% at 100 BeaverTails samples — a meaningful improvement. However, prefacing with 1000 GSM8K samples raises the LoX-defended model’s ASR to 63% at 100 harmful samples and 89% at 1000, nearly matching the undefended baseline. These results demonstrate that while LoX improves robustness against direct attacks, benign fine-tuning can substantially undermine this protection when used as a precursor to harmful adaptation.

1 Motivation & Problem Statement

1.1 Motivation

The democratization of Large Language Models (LLMs) through open-weight releases has introduced a critical security paradox. While models like Llama-2-7B-Chat are released with extensive safety alignment, this protection is not an immutable property of the model after deployment. Once the model weights are released, downstream users can fine-tune the model in ways that the original provider can no longer monitor or control. This makes post-release robustness an important security concern for open-weight models.

Prior work has shown that safety alignment can be sensitive to continued fine-tuning, even when the training corpus is benign and does not contain explicitly harmful content [6, 9].

This phenomenon is closely related to catastrophic forgetting: the model may lose or weaken previously learned safety behavior after being trained on new data. In our setting, this means that a model can begin to forget parts of its refusal behavior even before any harmful fine-tuning data is introduced.

This creates a practical attack risk. If benign fine-tuning can weaken safety alignment, then an attacker may be able to use benign data as a preparatory stage before introducing harmful examples. The first stage may appear harmless, while still making the model more vulnerable to a later harmful fine-tuning stage. Our project studies whether this sequential attack path can weaken a model’s safety behavior, and whether LoX remains effective under this setting.

1.2 Problem Statement

While safety alignment provides an essential guardrail for deployed LLMs, its tendency to degrade under continued fine-tuning demands robust defensive countermeasures [3, 6]. One recently proposed solution is LoX (Low-Rank Extrapolation), a pre-fine-tuning defense that identifies and amplifies safety-relevant low-rank directions in a model’s parameter space before the model is released [11]. The goal of LoX is to make safety alignment more resistant to downstream fine-tuning attacks.

The original LoX evaluation primarily considers direct harmful fine-tuning, where the attacker immediately trains the model on harmful examples. This work investigates whether LoX remains effective under a more subtle threat: a two-stage sequential fine-tuning attack that uses catastrophic forgetting as a preparatory mechanism. Rather than targeting safety alignment directly at first, the attacker begins by fine-tuning the model on benign data. After this first stage, the attacker fine-tunes the model on harmful data to test whether the model is now easier to push toward unsafe responses.

To model this threat, we use GSM8K as our benign fine-tuning corpus and BeaverTails as our harmful fine-tuning source [2, 7]. We evaluate the full attack pipeline against both an undefended Llama-2-7B-Chat model and a LoX-hardened

model. Safety degradation is measured using Attack Success Rate (ASR) on AdvBench harmful prompts, with Llama-Guard-3-8B used as the safe/unsafe judge.

Our goal is to determine whether catastrophic forgetting represents a blind spot in LoX’s threat model. Specifically, we ask whether a benign fine-tuning stage can make later harmful fine-tuning more effective, and whether LoX still preserves safety alignment under that sequential attack. Our final results suggest that LoX improves robustness compared to the undefended chat model, but does not fully prevent safety degradation under stronger harmful fine-tuning or sequential attack settings.

1.3 Threat Model

Attacker Model: We model an adversary with complete access to the weights of a publicly released, safety-aligned open-weight model. The attacker has the computational resources necessary to perform arbitrary fine-tuning, such as LoRA-based adaptation, and has access to two datasets: a benign corpus used in the first attack stage and a harmful dataset used in the second. For the benign stage, we use GSM8K [2], a dataset of grade-school mathematics word problems requiring multi-step arithmetic reasoning. We selected GSM8k because its math-focused content is semantically unrelated to harmful instruction following.

The attacker’s primary objective is to produce a model that complies with harmful prompts that the original safety-aligned model would likely refuse. We measure this objective using Attack Success Rate (ASR) on AdvBench harmful prompts. The attacker is not assumed to know the internal safety-critical subspaces identified by LoX; the attack only relies on downstream fine-tuning access.

Defender Model: The defender is the original model provider. Prior to release, the defender applies the LoX hardening procedure to the safety-aligned model, modifying its weights to increase robustness against fine-tuning attacks. After release, the defender has no visibility into how downstream users fine-tune the model and cannot constrain future fine-tuning pipelines. This assumption reflects the realistic constraints of open-weight deployment, where the provider gives up direct control once the weights are public.

The defender’s goal is to ensure that the model’s safety alignment is preserved even after downstream fine-tuning. In particular, a successful defense should reduce the production of policy-violating outputs regardless of whether the attacker targets alignment directly through harmful fine-tuning or indirectly through an initial benign fine-tuning stage.

2 Related Work

2.1 Safety Defenses in Open-Weight Models

Modern open-weight LLMs are typically safety-aligned using supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), or related preference-optimization techniques such as Direct Preference Optimization (DPO). These approaches modify a pretrained base model to reduce harmful or policy-violating outputs while preserving general capability. However, a growing body of work has shown that such alignment is not always deeply integrated into model representations; rather, it can occupy a relatively small and fragile region of parameter space that may be disrupted by later fine-tuning updates [10].

After the risks of fine-tuning-based alignment degradation were identified, several categories of mitigation strategies emerged. One line of work focuses on defenses applied before downstream fine-tuning occurs, such as modifying or strengthening the aligned model so that it is more robust to later updates [13]. Another line targets the fine-tuning stage directly by constraining updates, retaining base model features, or incorporating adversarial training to reduce safety degradation [15]. A third category focuses on post-release recovery, restoring safety properties in models that have already been fine-tuned [4].

LoX (Low-Rank Extrapolation) fits directly into the first category: a pre-fine-tuning defense applied after alignment, but before downstream users modify the model. It requires only an aligned model checkpoint and its corresponding base model.

2.2 Fine-Tuning Attacks on Aligned Models

A related line of work examines the vulnerability of aligned LLMs to post-deployment fine-tuning attacks. These attacks typically assume that an adversary obtains access to model weights and performs additional fine-tuning to induce harmful behavior.

The most straightforward attack strategy involves direct harmful fine-tuning, in which an adversary trains the model on an explicitly adversarial dataset designed to elicit policy-violating outputs. This approach has been shown to be effective even at relatively small scales; a limited number of harmful samples can substantially degrade alignment in models that underwent safety training [6]. Importantly, prior work also suggests that even fine-tuning on benign or domain-specific tasks can unintentionally undermine safety alignment.

A second class of attacks focuses on backdoor insertion during supervised fine-tuning. Here, the adversary embeds hidden triggers into the training data such that the model behaves safely under normal conditions but produces harmful outputs when a specific trigger is present. These attacks are particularly concerning because the backdoor can persist

through subsequent fine-tuning rounds, making them difficult to detect or remove after the fact [3].

A third approach exploits low-rank adaptation methods such as LoRA to minimize computational cost while still achieving substantial behavioral shifts [12]. Because LoRA updates only a subset of trainable parameters, an attacker can redirect model behavior with far fewer resources than full fine-tuning requires. This lowers the barrier to entry for adversaries and makes fine-tuning attacks more practical in resource-constrained environments.

However, most existing evaluations focus on single-stage attacks in which the adversary directly optimizes toward harmful outputs. Our work extends this threat model by explicitly evaluating a two-phase attack: benign fine-tuning to weaken safety behavior through catastrophic forgetting, followed by harmful fine-tuning to redirect behavior. This exposes a gap in how existing defenses, including LoX, have been evaluated.

2.3 Catastrophic Forgetting in Large Language Models

Catastrophic forgetting has long been studied in continual learning and neural network research. In the context of LLMs, it has been observed that fine-tuning on domain-specific or task-specific data can degrade performance on unrelated tasks [8]. More recently, researchers have identified safety alignment as a behavior that can also be vulnerable to such degradation [1].

Unlike traditional continual learning settings, safety alignment may correspond to relatively low-rank or low-magnitude parameter modifications layered on top of a powerful pre-trained base model. Consequently, later fine-tuning updates can disproportionately affect safety-related behavior, even when the fine-tuning data itself is benign. In our project, this is the key risk: benign fine-tuning may not directly teach the model unsafe behavior, but it may weaken the refusal behavior that prevents unsafe outputs.

Prior work has documented safety drift under fine-tuning, but systematic evaluation of defenses under catastrophic-forgetting-based attack preparation remains limited [12]. Our contribution is to treat catastrophic forgetting as an active adversarial mechanism rather than only an incidental training failure. We use benign GSM8K fine-tuning as a preparatory stage before harmful BeaverTails fine-tuning, then evaluate whether LoX remains robust under that sequential setup.

2.4 Low-Rank Defenses and the LoX Framework

Low-rank parameterization techniques, such as LoRA, have been widely used to reduce the computational cost of fine-tuning [5]. More recent defenses, such as SaLoRA, also leverage the observation that alignment updates often exhibit low-rank structure and propose constraining or regularizing these

updates to preserve safety-relevant directions in parameter space [10].

Building on this perspective, the Low-Rank Extrapolation (LoX) framework introduces a training-free defense that amplifies alignment-relevant components of a model. Rather than modifying the downstream fine-tuning procedure, LoX operates before downstream fine-tuning by decomposing the alignment update, ΔW_{align} , into its principal components using singular value decomposition and extrapolating along its top-ranked directions. These directions correspond to the most salient safety-related subspaces learned during alignment. By reinforcing them, LoX increases the model’s resistance to subsequent fine-tuning that might otherwise degrade safety behavior.

Formally, LoX constructs an updated parameter set by augmenting the aligned model with a scaled projection of ΔW_{align} onto its top- k singular components, where k is chosen as the minimum rank required to preserve safety performance under an Attack Success Rate criterion. This formulation isolates and strengthens the most safety-critical structure in the alignment update while avoiding unnecessary parameter modifications.

Empirical results reported in the LoX work show that this approach improves robustness against direct adversarial fine-tuning compared to unprotected models. However, these evaluations primarily focus on direct or single-stage attack settings. Our work evaluates whether the same low-rank defense remains effective when the attacker first applies benign fine-tuning before harmful fine-tuning.

3 Research Questions

We organize our experiments around three research questions:

Q1: Does LoX work against direct harmful fine-tuning?

We compare the LoX-defended model against the undefended chat model as the number of BeaverTails harmful samples increases. This establishes a baseline performance for the LoX defense for which to compare our iterative attack.

Q2: Does catastrophic forgetting weaken LoX?

We test whether benign GSM8K fine-tuning before harmful BeaverTails fine-tuning produces higher ASR than direct harmful fine-tuning alone. This iterative fine-tuning attack serves as the main focus of this paper.

Q3: How sensitive is LoX to fine-tuning hyperparameters?

We vary the learning rate during harmful fine-tuning to test whether stronger gradient updates can overcome the defense. Since LoX works by flattening the gradients around safety-relevant subspaces of the models’ parameters, we test whether fine-tuning with a higher learning rate overcomes the defense.

4 Methodology

Our evaluation follows a sequential fine-tuning pipeline designed to measure how safety alignment changes under different attack conditions. For each experiment, we begin with either the original Llama-2-7B-Chat model or a LoX-defended variant of the same model. We then apply one or more stages of downstream fine-tuning before evaluating the resulting model on harmful prompts.

For direct harmful fine-tuning experiments (Q1), models are fine-tuned only on BeaverTails harmful samples. We vary the number of harmful training examples to measure how rapidly safety behavior degrades under increasingly strong attacks.

For sequential fine-tuning experiments (Q2), models first undergo benign fine-tuning on GSM8K before being fine-tuned on BeaverTails. This setup evaluates whether benign adaptation weakens safety alignment prior to harmful training. We compare ASR under direct and sequential fine-tuning to measure the extent to which the benign stage increases susceptibility to harmful adaptation.

For hyperparameter sensitivity experiments (Q3), we vary the learning rate during harmful fine-tuning while keeping the dataset size fixed. This isolates the effect of update magnitude on the robustness of the LoX defense.

After fine-tuning, all models are evaluated using harmful prompts from AdvBench. Generated responses are classified as safe or unsafe using Llama-Guard-3-8B, and safety degradation is measured using Attack Success Rate (ASR).

5 Experimental Setup

Models. We use Llama-2-7B as the base model [14], along with its safety-aligned checkpoint. We generate the LoX-defended model by applying the LoX procedure to the aligned model. This produces a hardened model intended to better preserve safety alignment after downstream fine-tuning. Llama-Guard-3-8B is used as the evaluation classifier for ASR measurements.

Datasets. GSM8K [2] is used as the benign fine-tuning dataset. It contains grade-school math word problems and is semantically unrelated to harmful instruction following. BeaverTails [7] is used as the harmful fine-tuning dataset. AdvBench [17] harmful prompts are used exclusively for evaluation, ensuring separation between attack training data and evaluation prompts.

Fine-tuning setup. Our main experiments use LoRA-based fine-tuning implemented with Unsloth [16]. In addition to the standard LoRA adapters, we unfreeze the language modeling head (`lm_head`), the final layer that maps hidden states to next-token logits. This increases the percentage of trainable

parameters from approximately 4.53% to 6.39%, providing additional adaptation capacity while remaining substantially more efficient than full-parameter fine-tuning. Unless otherwise stated, experiments use a learning rate of 5×10^{-4} .

ASR evaluation. We measure safety degradation using Attack Success Rate (ASR). For each harmful prompt, the target model generates a completion that is then classified as safe or unsafe by Llama-Guard-3-8B. ASR is computed as:

$$ASR = \frac{\# \text{ unsafe responses}}{\# \text{ total prompts}} \times 100$$

Lower ASR indicates stronger refusal behavior, while higher ASR indicates increased production of unsafe responses.

6 Evaluation & Results

6.1 Judge Baseline Evaluation

Before relying on ASR we first evaluate the judge model itself. This performance is likely because the safety annotations in the BeaverTails dataset are calibrated to flag only outright refusal to answer a harmful prompt as “safe”, while many of the sample responses reframe a harmful prompt into a safer connotation (but are still marked as ‘unsafe’). However, Llama-Guard-3-8B performed strongly on the AdvBench harmful samples, with 99.0% accuracy, 2.0% false-negative rate, and 0.0% false-positive rate over 100 samples. Since we use the AdvBench dataset in our ASR evaluations, we can be confident in the results.

Test Set	Samples	FNR	FPR	Accuracy
AdvBench	100	2.0%	0.0%	99.0%
BeaverTails	1000	47.6%	4.0%	74.2%

Table 1: Llama-Guard-3-8B judge baseline evaluation.

6.2 Evaluations & Findings

We report ASR results across three experimental conditions corresponding to each of our three research questions. All evaluations use AdvBench harmful prompts judged by Llama-Guard-3-8B.

Q1: LoX vs. Undefended Model Under Direct Harmful Fine-Tuning. Figure 1 shows ASR as a function of BeaverTails sample count for both the undefended chat model and the LoX-defended model under direct harmful fine-tuning, with no GSM8k pre-training stage.

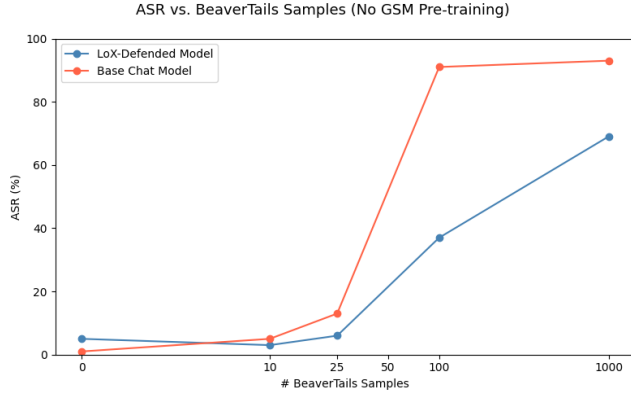


Figure 1: ASR vs. BeaverTails sample count for undefended and LoX-defended models under direct harmful fine-tuning.

Both models remain robust at low sample counts, with ASR near 0% up to approximately 25 BeaverTails samples. Beyond this threshold, both models experience a sharp increase in ASR. The undefended chat model reaches 91% ASR at 100 samples and 93% ASR at 1000 samples. The LoX-defended model reaches 37% ASR at 100 samples and 69% ASR at 1000 samples.

These results confirm that LoX consistently reduced ASR relative to the undefended model under direct harmful fine-tuning. At 100 BeaverTails samples, the LoX defense reduces ASR by approximately half relative to the undefended model. At 1000 samples, the gap narrows but remains substantial. Nevertheless, LoX does not fully prevent safety degradation under this attack, particularly as the number of harmful fine-tuning samples grows.

Q2: Iterative Fine-Tuning Attack via Catastrophic Forgetting. Figures 2a and 2b compare ASR with and without a 1000-sample GSM8k benign pre-training stage, for the undefended chat model and the LoX-defended model respectively.

The benign fine-tuning stage substantially weakens both models prior to harmful fine-tuning. For the LoX-defended model, introducing 1000 GSM8K samples before harmful fine-tuning raises ASR to 63% at 100 BeaverTails samples and 89% at 1000 BeaverTails samples. This is in stark contrast to 37% and 69% respectively under the direct attack. At 100 BeaverTails samples, this represents nearly a doubling of ASR. At 1000 samples, the sequential attack brings the LoX-defended model to 89% ASR, representing near full safety-alignment collapse.

These results are consistent with our hypothesis that catastrophic forgetting contributes to increased vulnerability during subsequent harmful fine-tuning. *Even though the GSM8K fine-tuning stage contains no harmful content, it appears to erode safety-relevant parameter structure enough to make the model significantly more susceptible to subsequent harmful fine-tuning. LoX retains some residual robustness even in the*

sequential setting, but the defense is substantially weakened relative to the direct attack baseline.

Q3: Learning Rate Sensitivity of the LoX Defense. Figure 3 shows ASR for the LoX-defended model at 50 BeaverTails samples across a range of learning rates from 1×10^{-5} to 5×10^{-4} .

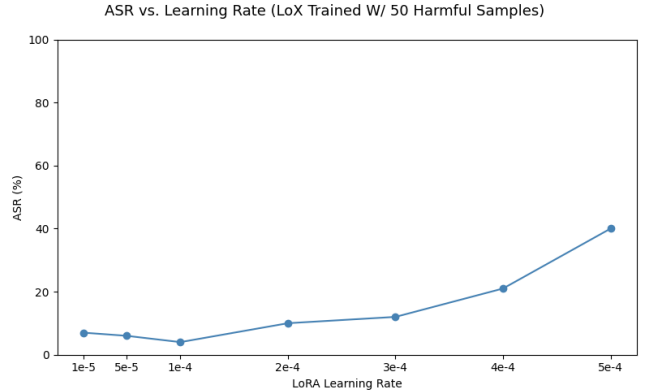


Figure 3: ASR of the LoX-defended model at 50 BeaverTails samples across learning rates.

ASR remains low and relatively stable across most of the tested learning rate range, suggesting that the LoX defense is reasonably robust to moderate variation in fine-tuning step size. However, a clear upward trend emerges at higher learning rates, with meaningful safety degradation becoming apparent near the upper end of the range. At 5×10^{-4} , the LoX-defended model reaches 38% ASR on only 50 BeaverTails samples.

This finding is consistent with the design intent of LoX: the defense amplifies safety-relevant directions in parameter space, effectively increasing the gradient magnitude required to overwrite them. Larger learning rate updates produce parameter updates large enough to substantially alter the reinforced safety directions. *Taken together with the sequential fine-tuning results from Q2, these results suggest that LoX’s robustness has identifiable limits that an informed attacker could exploit through either stronger gradient updates or a preparatory forgetting stage.*

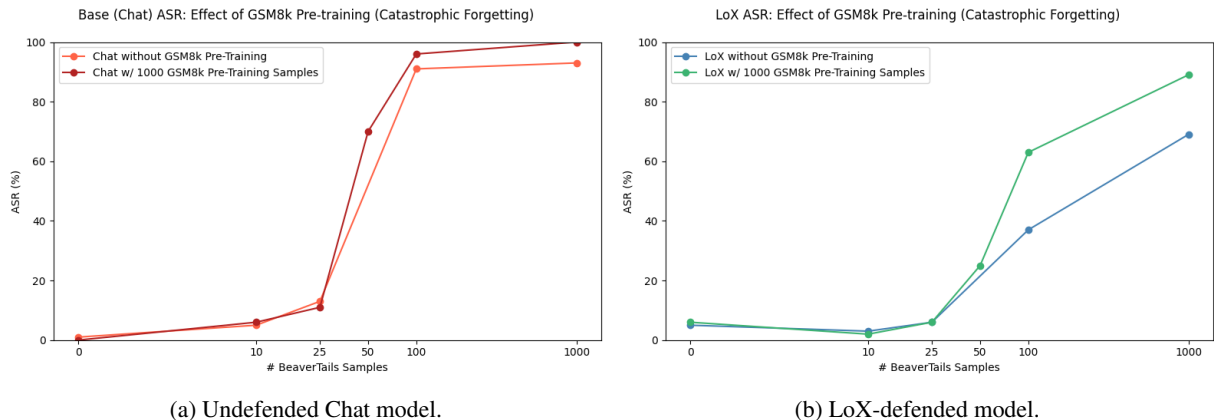


Figure 2: Effect of iterative fine-tuning (GSM8k pre-training with 1000 samples, then BeaverTails samples) on the two models.

Taken together, these results show that LoX improves robustness against direct harmful fine-tuning, but its effectiveness decreases under stronger attack conditions. Higher learning rates increase ASR during harmful adaptation, and sequential fine-tuning with benign GSM8K data further weakens the defense prior to harmful training. Although LoX consistently reduces ASR relative to the undefended model, the sequential fine-tuning setup substantially narrows this gap, suggesting that benign fine-tuning may increase susceptibility to subsequent harmful adaptation.

7 Limitations and Future Work

Judge reliability. Our ASR evaluation depends on Llama-Guard-3-8B. The judge performs well on AdvBench, but its lower accuracy and high false-negative rate on BeaverTails show that automated judging is imperfect. Some harmful responses may be missed, and ASR should be interpreted as an approximate safety signal rather than a perfect measurement.

Model specificity. We evaluate Llama-2-7B-Chat and a LoX-defended variant. Results may not generalize to larger models, different model families, or models aligned with different procedures.

Narrow attack surface. Our benign fine-tuning dataset is GSM8K and our harmful fine-tuning dataset is BeaverTails. Other benign domains, such as code generation or instruction following, may interact differently with safety alignment. Other harmful datasets may also produce different ASR patterns.

Low-rank fine-tuning setup. Our main attack uses LoRA with the language modeling head unfrozen. This is more practical than full-parameter fine-tuning, but it is not identical to a full fine-tuning threat model.

Future work. Several directions remain open. First, our attack relies on LoRA with a partially unfrozen model; testing LoX against full-parameter fine-tuning would establish a more permissive upper bound on attacker capability. Second, the hyperparameter space of our attack is only partially explored. LoRA rank, alpha, warmup steps, and layer selection all likely interact with how quickly safety alignment erodes, and a systematic evaluation might uncover more vulnerabilities in the defense. Third, all experiments use Llama-2-7B-Chat; whether the sequential forgetting attack generalizes to larger models or different alignment procedures is an open question. Finally, judge reliability remains a bottleneck; future evaluations should consider using a more robust judge, incorporating human review, or using prompt sets specifically designed to stress-test judge calibration.

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A Appendix

A.1 Data

1. Preliminary QLoRA Results

Model	Condition	Unsafe/Total	ASR
Llama-2-7B-Chat	Baseline	0/25	0.0%
Llama-2-7B-LoX	LoX baseline	2/25	8.0%
Chat-stage2-n10	10 GSM + 10 BT	0/25	0.0%
Chat-stage2-50	50 GSM + 50 BT	0/25	0.0%
LoX-stage2-100	100 GSM + 100 BT	1/25	4.0%

The ASR results after QLoRA finetuning show that weaker parameter-efficient fine-tuning did not produce strong catastrophic forgetting in our tested settings. This motivated a stronger LoRA/Unsloth setup with the language modeling head unfrozen.

2. Iterative Fine-Tuning Results Summary

Model	GSM Samples	BT Samples	ASR
Chat	0	0	1%
Chat	0	10	5%
Chat	0	25	13%
Chat	0	100	91%
Chat	0	1000	93%
Chat	1000	10	6%
Chat	1000	25	11%
Chat	1000	50	70%
Chat	1000	100	96%
Chat	1000	1000	100%
LoX	0	0	5%
LoX	0	10	3%
LoX	0	25	6%
LoX	0	100	37%
LoX	0	1000	69%
LoX	1000	10	2%
LoX	1000	25	6%
LoX	1000	50	25%
LoX	1000	100	63%
LoX	1000	1000	89%

The above table summarizes the main LoRA/Unsloth ASR results across various combinations of GSM8k and Beaver-Tails samples. Without GSM8K pre-training, the undefended chat model becomes unsafe quickly as the number of Beaver-Tails samples increases. The LoX-defended model degrades more gradually.

3. Hyperparameter ASR Evaluations

Learning Rate	ASR
1×10^{-5}	7%
5×10^{-5}	6%
1×10^{-4}	4%
2×10^{-4}	10%
3×10^{-4}	12%
4×10^{-4}	21%
5×10^{-4}	38%

In these experiments, the LoX model is fine-tuned with 50 harmful BeaverTails samples while varying the learning rate. We see that ASR stays relatively low across most of the tested learning-rate range, but rises sharply at 5×10^{-4} . This suggests that LoX robustness depends not only on the model, but also on the strength of the downstream fine-tuning configuration.

A.2 Statement of Work

- **Jorge Manuel Torre:** Helped implement the experimental pipeline, generated valid ASR result CSVs, organized shared evaluation outputs, contributed to the methodology and results analysis, and helped prepare the final presentation.
- **Elias Mengistu:** Contributed to dataset selection, evaluation framing, benign fine-tuning discussion, and project presentation content.
- **Alex Levy:** Abstract, threat model, related work, methodology, limitations, future work, fine-tuning scripts, graphs, baseline evaluations, data & results.
- **Jeevan Shahi:** Contributed to final report and presentation documentation.

A.3 Usage of AI

AI was used in our project to make the grammar and layout of this proposal more professional. Specifically, we gave AI our rough draft and implemented its improvements and suggestions to produce the final paper. We used the Chat-GPT5 and Claude Sonnet/Opus 4.6 models.